

Audio Engineering & Software Portfolio

Audio Software, DSP & Interactive Systems — C++/JUICE, MATLAB & Python

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1 Professional Profile

Software developer focused on **Digital Signal Processing (DSP)** and professional audio tooling, with an ecosystem of VST3 plugins published under the **ESP** brand and distributed through a self-built web store. Experienced in modern **C++** with the **JUCE** framework, **MATLAB** and **Python** — from wavetable synthesis and real-time dynamics processing to gesture-controlled instruments with computer vision, embedded systems (ESP32, Raspberry Pi/ROS2) and applied machine learning. Prioritises *lock-free* architectures, algorithmic efficiency and clear real-time data visualisation.

2 Audio Plugins — C++ / JUCE

VST3 / Standalone plugins developed in C++ with JUCE 8, published under the **ESP** brand and distributed at [ESP Plugin Store](#).

2.1 SYNTH/1 — Polyphonic Wavetable Synthesizer

16-voice polyphonic synthesizer with a Serum-inspired architecture: wavetable morphing, a unison engine, a step sequencer and a full effects rack.

- **Oscillator:** 4 band-limited wavetables with continuous morphing and 4 phase-warp modes (Sync, Bend, PWM); real-time waveform visualiser. The bank is **mip-mapped** — 21 half-octave bands per waveform over 2048-sample tables, each holding only the harmonics that stay below Nyquist (up to 256) — read with 4-point cubic Hermite (Catmull–Rom) interpolation on a double-precision phase, and cross-faded between adjacent bands so no step is audible at a band boundary.
- **Voice architecture:** up to 8-oscillator unison with constant-power detune and stereo spread; Poly/Mono/Legato voice modes with per-voice multiplicative glide; chain: oscillator → pan → tanh drive → SVT filter → ADSR.
- **Modulation & sequencer:** LFO routable to cutoff and pitch with a mod-depth ring drawn around the knob; step sequencer of up to 16 steps with internal clock or DAW triggering, fully serialised in the plugin state.
- **FX & EQ:** chorus, phaser, tempo-synced delay, tanh saturation with 2× oversampling and reverb; **interactive** 3-band EQ (drag the bands directly on the frequency response) with a live FFT spectrum overlay.
- **Engineering:** lock-free audio↔UI communication (atomics and ring buffers), XML presets via APVTS, 7 colour themes with a custom *LookAndFeel*.
- **Measured:** an offline harness (`analysis/measure_oscillator.cpp`, compiled against a minimal JUCE stub) sweeps a sawtooth across the keyboard and reports the alias-to-signal ratio. Against the previous 64-harmonic, linearly-interpolated design the mip-mapped oscillator is **96 dB better at A4** and **139 dB better at C8** — and 4 dB *worse* below ~86 Hz, where the table image rather than aliasing sets the floor. That regression is characterised, not hidden.
- **Repository:** [GitHub – Synth 1.0](#)

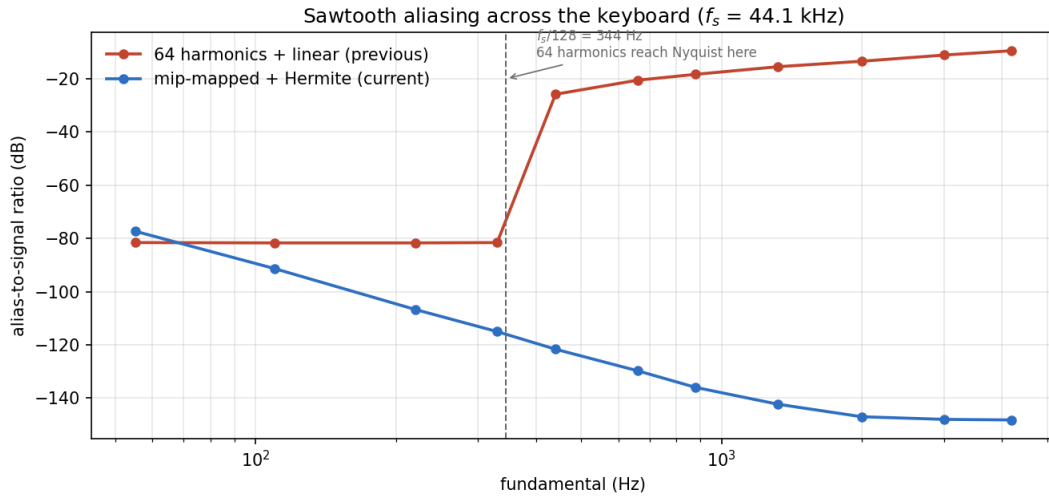


Figure 1: Measured sawtooth alias-to-signal ratio across the keyboard. The previous design collapses above $f_s/128$, where 64 harmonics reach Nyquist; the mip-mapped bank keeps falling. Below ~ 86 Hz it is 4 dB worse, set by the table image.

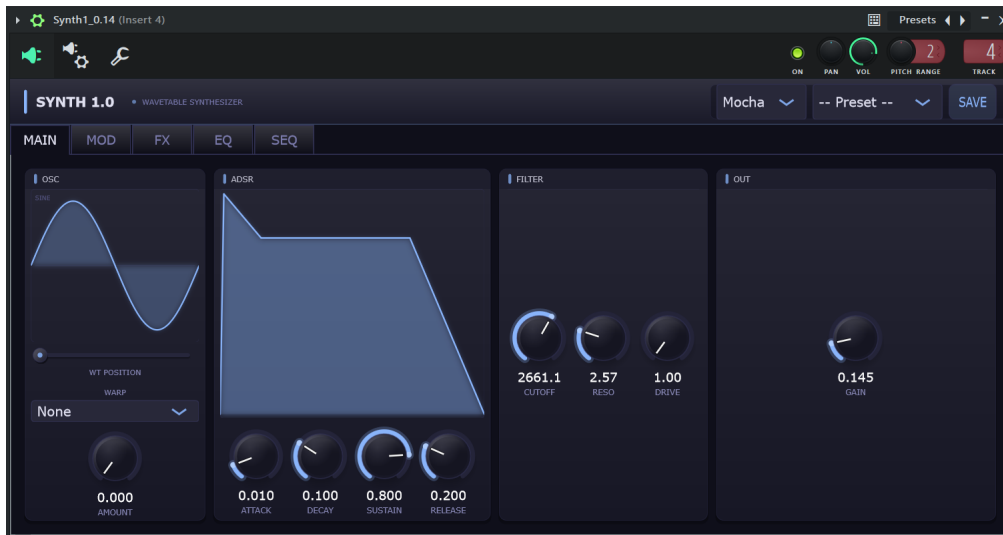


Figure 2: SYNTH/1: main tab with wavetable visualiser, filter and ADSR.

2.2 SPECTRUM — Real-Time Spectrum Analyzer

High-precision spectral analysis tool operating in real time with optimised latency.

- **DSP engine (FFT):** 2048-point FFT with Hann windowing; audio is pulled through *thread-safe* FIFO structures, avoiding any blocking of the high-priority audio thread.
- **Rendering:** spectral curve with Bézier interpolation and moving-average smoothing, logarithmic frequency mapping (20 Hz–20 kHz), 60 FPS refresh with a minimal CPU footprint.
- **Metering:** analog-style VU meters in dBFS with smooth falloff ballistics, stereo panning indicator and FREEZE function; multiple views (spectrogram, 3D waterfall, waveform) and colour themes.
- **Repository:** [GitHub – Spectrum Analyzer](#)

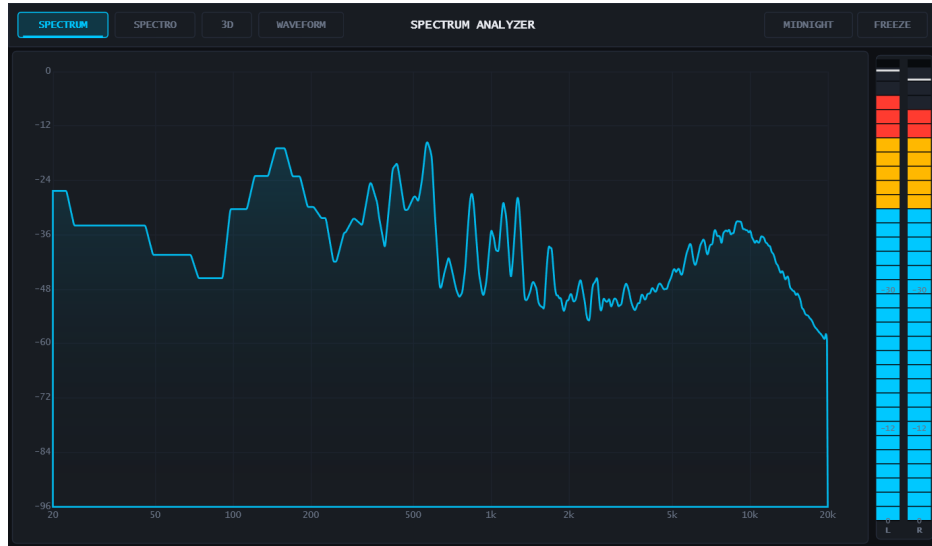


Figure 3: SPECTRUM showing the smoothed frequency response and VU meters.

2.3 VERTEX — Dynamic Range Compressor

Professional-grade dynamics processor with exhaustive control over the signal envelope and instant visual feedback of the gain reduction.

- **Architecture & APVTS:** full *AudioProcessorValueTreeState* integration for safe communication between the processing thread and the GUI, with persistent preset storage.
- **Dynamics processing:** gain computation with attack and release envelopes parameterisable in real time; Threshold, Ratio, Attack, Release and Makeup Gain controls.
- **Visual feedback:** dynamic transfer-curve (*knee*) display that updates with the user's parameters, complemented by asynchronous peak meters.
- **Repository:** [GitHub – VERTEX](#)



Figure 4: VERTEX: transfer curve and gain-reduction metering.

2.4 ESP-L1 — Brick-Wall Limiter

Instant-attack limiter with built-in spectral analysis to monitor the effect of the processing on the frequency content.

- **Limiting:** per-sample envelope detector with exponential decay (10–1000 ms release with perceptual skew), -60 to 0 dB threshold and post-limiting output gain.
- **Pre/post analysis:** two independent FFT pipelines (2048-point, Hann) draw the spectrum before and after limiting, overlaid, alongside VU and gain-reduction meters.
- **Threading:** audio/UI state shared through atomics; the UI computes FFTs and ballistics at 30 Hz without touching the audio thread.
- **Repository:** [GitHub – ESP-L1](#)



Figure 5: ESP-L1: pre/post limiting spectrum and gain-reduction meter.

2.5 MEGACRUSHER — Distortion, Saturation & Bit-Crusher

Distortion plugin with three saturation algorithms, bit-depth reduction and an animated interface that reacts to the signal.

- **Saturation:** SOFT (tanh), HARD (hard-clip) and FOLD (wave-folder that mirrors the signal back into range, generating complex harmonics) modes; $1\times$ to $40\times$ drive stage.
- **Timbre:** 16-bit down to 2-bit bit-crusher, one-pole tilt filter for tone control and dry/wet mix for parallel distortion.
- **Visualisation:** live transfer-curve display, stereo peak-hold VU with colour-coded segments and an animated UI (ember particles and flames reacting to the Drive level).
- **Repository:** [GitHub – MegaCrusher](#)



Figure 6: MEGACRUSHER: live saturation curve and stereo VU.

2.6 ESP Plugin Store — Web Distribution

Self-built web store distributing the ESP-brand plugins, deployed at esp-plugin-store.vercel.app.

- **Frontend:** self-contained React 18 SPA (no build step); seeded generative waveform art per plugin, screenshot carousel and a download modal with OS and format selection.
- **Distribution:** VST3 binaries packaged under the GPL v3 licence, SEO sitemap and continuous deployment on Vercel.

2.7 Basic Oscillator — Oscillator Generator

Three independent oscillators with waveform selection (sine, saw, square) built on `juce::dsp`; first project of the JUCE ecosystem and the experimental foundation for SYNTH/1.

3 Gestural Instruments — Python & Computer Vision

3.1 AirStems — Gestural Stem Remixing (Musicathon by Musixmatch Pro, 2026 — LALAL.AI Special Prize)

“Conduct and remix a song’s stems with your bare hands”: source separation + synced lyrics + gestural control over a live DSP engine.

Won the LALAL.AI Special Prize at this global online hackathon; since then **featured in a published LALAL.AI interview** — [“How AirStems uses the LALAL.AI API”](#) — and invited to write a **technical guest article** for the LALAL.AI blog and Dev.to.

- **Stems & lyrics:** vocals/drums/bass/other separation via the **LALAL.AI** API (with a local

Demucs fallback) and **Musixmatch** time-synced lyrics scrolling karaoke-style.

- **Gestural control:** two-hand MediaPipe tracking — right-hand fingers toggle each stem (with hysteresis), left-hand height drives a low-pass filter and finger spread the reverb; a fist performs a drop.
- **Real-time engine:** stereo multi-stem mixer on a `sounddevice` callback with gain changes quantised to the beat grid (librosa on the drum stem) and click-free per-block ramps.
- **Demo:** [30-second video demo \(YouTube\)](#)
- **Repository:** [GitHub](#) – [AirStems](#)

3.2 Aetheric Geometry — Real-Time Gesture Instrument

Musical instrument played with bare hands: a webcam feeds MediaPipe hand landmark detection and the geometry of the hands is translated into additive synthesis with just intonation.

- **Gesture pipeline:** gesture detectors (pinch, kiss, pyramid, cross, prayer, open palm) on top of a state machine (IDLE / THREAD / POLYGON) with debouncing and robust hand assignment; waveform selection by counting extended fingers.
- **Synthesis engine:** additive oscillators with just-intonation ratios, asymmetric envelope, IIR low-pass filter mapped to hand height, Schroeder reverb (4 comb filters) and tremolo driven by the polygon's aspect ratio.
- **Product:** built-in WAV recording, optional MIDI output, packaged as an executable (PyInstaller) and a *pytest* suite that requires no camera or audio hardware.
- **Demo:** [video demo \(YouTube\)](#)
- **Repository:** [GitHub](#) – [Aetheric Geometry](#)

3.3 AirPiano — Webcam Air Piano

8-note piano (plus a drum-kit variant) played in the air: frame-difference motion detection with OpenCV over on-screen key zones and sample triggering with pygame; motion-threshold calibration for the room's lighting.

4 Hardware & Embedded Systems

4.1 T-8 Groovebox — 16-Step Sequencer, PCB Design (KiCad, 2026)

A 16-step groovebox in the spirit of the Roland T-8, designed from scratch in KiCad: a 290×210 mm two-layer board with 98 components, roughly 18 m of copper across 2,574 track segments and 1,150 vias. The design is closed — DRC clean, fabrication package generated.

- **Architecture:** ESP32-WROOM-32 driving two MCP23017 expanders (16 step pads on 0x20, 8 function keys on 0x21); a CD74HC4067 multiplexes 10 potentiometers onto a single ADC pin; 16 WS2812B LEDs behind a 74AHCT1G125 level shifter so the 5 V data line is driven correctly from 3.3 V logic.
- **Audio:** one I²S bus feeding two slave DACs in parallel — a PCM5102 for line out and a MAX98357A class-D module for the speaker, its `SD_MODE` pin held at 1.04 V by a stiff 10 k/2.7 k divider to force the $(L + R)/2$ mono mix.
- **I/O:** MIDI in and out on 3.5 mm TRS Type A (the standard the real T-8 uses), USB-C power and flashing through a CH340C with a transistor auto-reset circuit, and a 2.4" ST7789 display.
- **Layout driven by playability:** pads on a 16 mm finger pitch, potentiometers at 26 mm centred above them, and every decoupling capacitor placed 4–5 mm from the supply pin it serves — two

of them moved to the bottom side because the ESP32 antenna keep-out puts the 3V3 and EN pads out of reach from above.

- **Integrity:** ground pour on both layers tied by 417 stitching vias, which closed 33 broken ground connections; 6 M3 mounting holes; 0.3 mm minimum drill; **0 DRC errors and 0 unrouted nets.**
- **Two defects worth the audit:** the amplifier carried the 17-pin raw-chip symbol on a 7-pad module footprint, leaving OUTP/OUTN and BCLK/LRCLK with no pads at all — the board would never have made a sound, and neither ERC nor the netlist can see it, since both work from the symbol. Separately, several footprints share one pad number across multiple physical pads (the tact switches have pads 1,1,2,2; the ESP32 thermal pad is split into eight) and the net had been assigned to only the first pad of each group, which was the real cause of 88 apparently orphaned tracks and every short reported against the thermal pad.

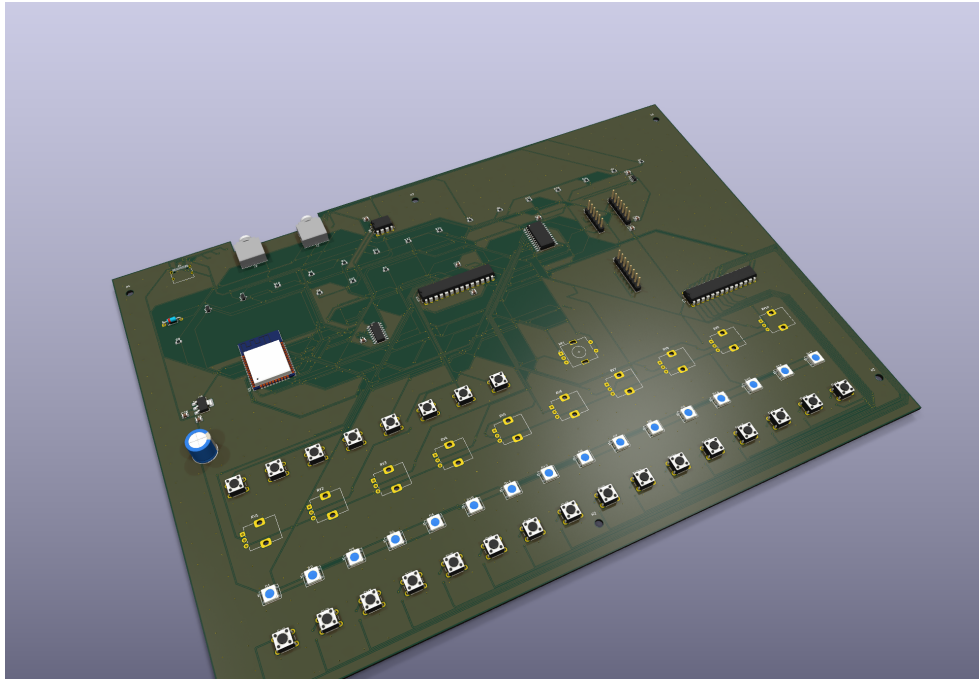


Figure 7: T-8 Groovebox: 3D render of the finished 290×210 mm board.

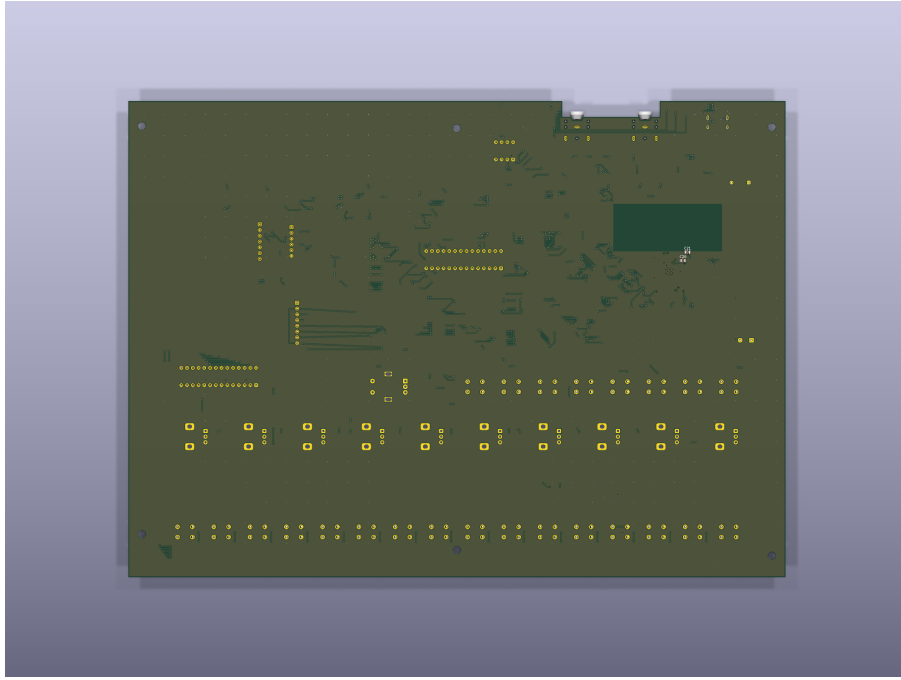


Figure 8: Bottom side: ground pour with 417 stitching vias. The dark patch is the ESP32 antenna keep-out, deliberately free of copper.

4.2 EDISA Inventory Drone — Autonomous Warehouse UAV (UPC PAE × EDISA, 2025–2026)

Autonomous drone that flies through a warehouse, builds a real-time 3D map, detects barcodes and packages, and generates a geolocated inventory — no manual operation. Team project of 14 students for the industry client EDISA; personal focus on WP2: the Ground Control Station and drone-control stack.

- **Onboard system:** Pixhawk/ArduCopter flight controller plus Raspberry Pi 5 running ROS2 Jazzy; Unitree 4D LiDAR feeding *Point-LIO* SLAM (~800,000 points per scan), Pi Camera, barcode detection (OpenCV, pyzbar, YOLOv8) with each scan tagged with its SLAM x/y/z position.
- **Ground Control Station:** cross-platform Electron app — telemetry/arming dashboard, artificial-horizon navigation, 3D point-cloud SLAM viewer and an inventory tab backed by SQLite with CSV/Excel export; talks to the drone over rosbridge WebSocket and MJPEG streaming, with no ROS install required on the laptop.
- **Autonomy:** mission-coordinator state machine (brain node) driving waypoint navigation over MAVLink v2; physics-based flight simulator and Monte Carlo mission simulations for validation.
- **Demo:** [Ground Control Station video demo \(YouTube\)](#)
- **Repository:** [GitHub – drone-gcs](#)

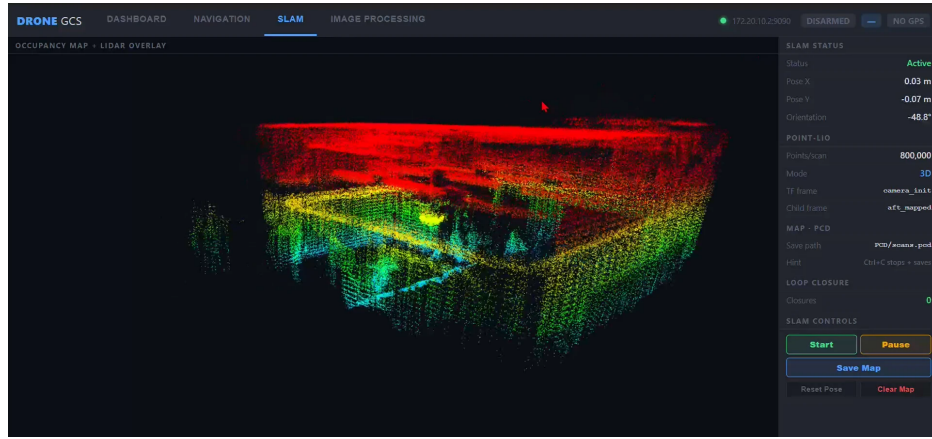


Figure 9: EDISA drone GCS: live 3D LiDAR point cloud in the SLAM viewer.

4.3 LaserHarp — Laser Harp

Musical instrument in which interrupting laser beams triggers the notes; the project covers both the electronics design and the control software.

- **Hardware:** Fritzing protoboard and PCB designs iterated v1 → v3 with ToF (*Time-of-Flight*) distance sensors and a Raspberry Pi control layer; ToF synthesizer prototype (“ToF-Synth”).
- **Software:** Python/Tkinter simulator of the instrument’s 16-character LCD menu (state machine for navigating and editing ADSR, 3-band EQ, effects and MIDI configuration) with Unicode-rendered value bars.
- **Repository:** [GitHub – ToF-Synth](#)

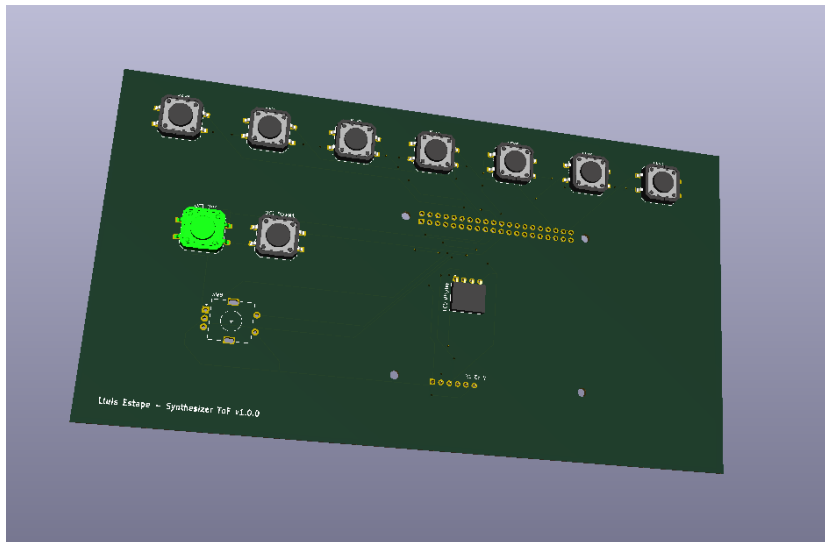


Figure 10: LaserHarp: 3D render of the ToF synthesizer PCB (v1.0.0).

4.4 ESP32 Cyberdeck (in progress)

Hand-built audio handheld (synth / sampler / MIDI): a sandwich of two perfboards with the controls on the walls of the central cavity.

- **3-microcontroller architecture:** ESP32-DevKitC as the brain (ILI9341 TFT display over SPI, I²S audio, I²C sensors), a second ESP32 as an I²C slave dedicated to reading 2 encoders, a slider, buttons and a D-pad, and an ESP32-CAM streaming MJPEG video over WiFi.

- **Audio chain:** I²S → PCM5102A DAC → PAM8403 amplifier → speaker; environmental sensors (AHT10, PIR) behind a TCA9548A multiplexer; 2000 mAh LiPo power with a charger/boost module.
- **Enclosure:** parametric case generator (10 bodies, display and camera windows, speaker grille, “hat” push-buttons) written in Python against the Autodesk Fusion 360 API.

5 Room Acoustics & Simulation

5.1 Theatre Design & Cinema Conversion — EASE 4.4 (Acoustics & Electroacoustics, UPC)

Complete acoustic design of a 675-seat theatre, modelled from scratch in **EASE 4.4** and then re-conditioned for use as a cinema. Team project with Pablo Sánchez; the deliverable is a report covering geometry, material treatment and the full objective validation chain.

- **Geometry:** circular-segment plan inspired by Greek theatres, two raked levels (390 + 285 seats at 0.55×0.95 m pitch), 114 vertices and 261 polygonal faces. Volume $9,092.7 \text{ m}^3$ over $4,787.9 \text{ m}^2$ of surface — $12.18 \text{ m}^3/\text{seat}$ once the 864 m^3 stage is removed, well above the $4\text{--}6 \text{ m}^3/\text{seat}$ target and therefore the constraint that drove the whole absorption strategy.
- **Material treatment:** frequency-dependent coefficients (GYP 2X5/8, WOOD, DIFFRACTAL) with the audience modelled as absorption blocks, plus a custom *Reflector_panel* material ($\alpha \approx 0.05$, scattering ≈ 0.10) on a 91.35 m^2 free-floating two-fold reflector above the stage, aimed at the rear rows.
- **Reverberation time:** Eyring RT held inside the $0.8\text{--}1.2$ s target across $125 \text{ Hz--}4 \text{ kHz}$ — 1.04 s at 125 Hz , 0.69 s at 500 Hz and 0.92 s at 4 kHz , with a mean absorption coefficient of 0.27 and a 7.60 m mean free path.
- **Echograms:** mirror-image impacts (order 4, 100 ms cut-off, 40 dB loss) at seats 2, 4, 6 and 7 in the 500 Hz , 1 kHz and 2 kHz bands; every arrival falls in Haas regions A–C, with region D empty — no echoes or acoustic shadowing.
- **Speech intelligibility:** STI mapping over the whole audience under an NC-25 noise floor. The stalls clear the $\text{STI} \geq 0.6$ requirement, while the rear balcony drops to $\approx 0.55\text{--}0.59$, traced to the excess volume per seat rather than to the treatment. The cinema reconfiguration (lower level only, reinforced electroacoustic system) lifts the minimum to **0.64** with uniform coverage.
- **Auralisation:** local ray tracing (order 6, 200 ms) through the EASE Ears module produced a dummy-head binaural impulse response for seat 2 (44.1 kHz , 1.70 s , AURA phase compensation), convolved with anechoic speech for subjective assessment; seats 3 and 4 on the upper level were auralised as a cross-check and match the STI map.

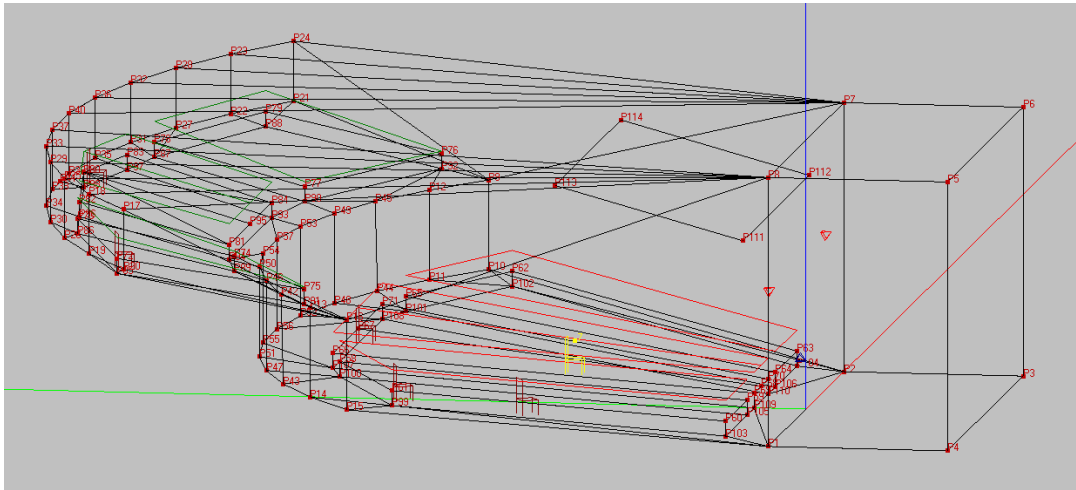


Figure 11: EASE 3D model of the theatre: two raked levels in a circular segment, with the stage volume on the right.

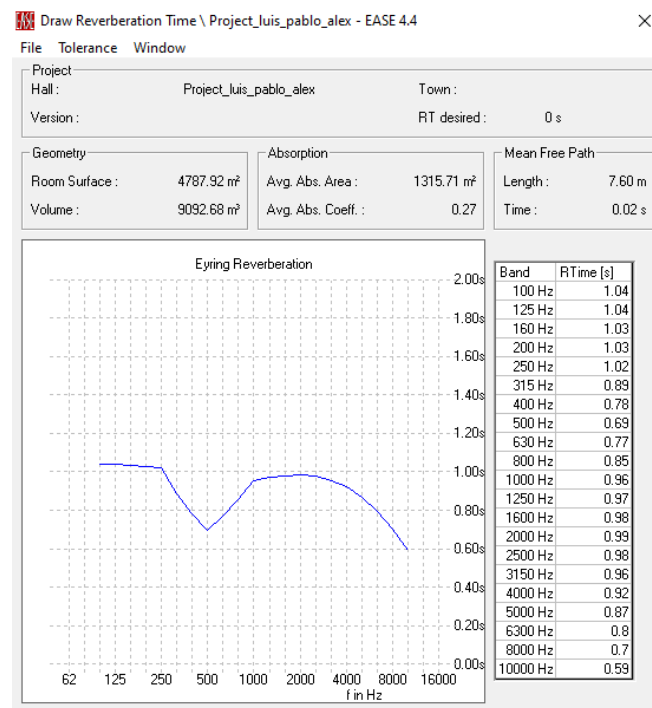


Figure 12: Eyring reverberation time per band against the 0.8–1.2 s design target.

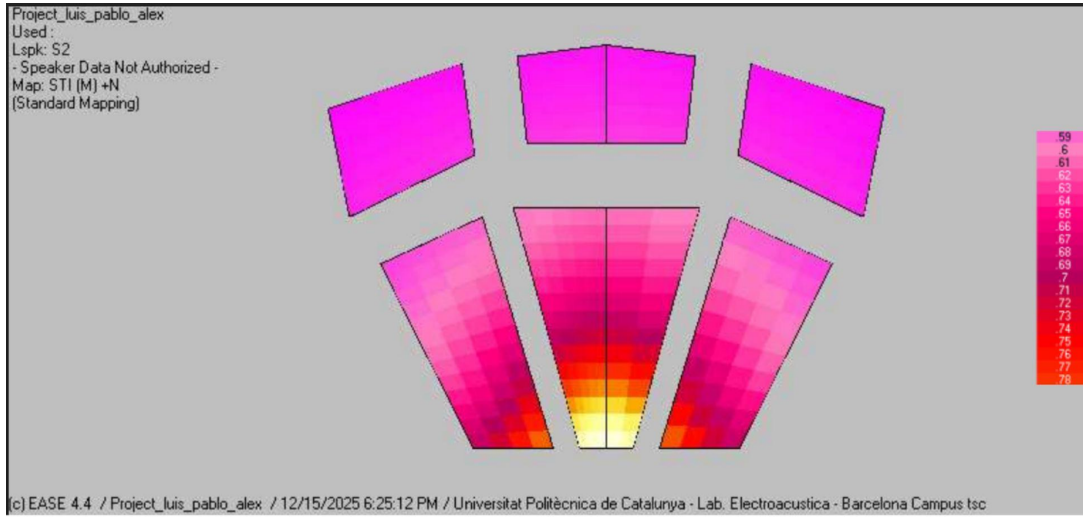


Figure 13: STI map over the audience areas under NC-25 background noise: high intelligibility in the front stalls, falling towards the rear balcony.

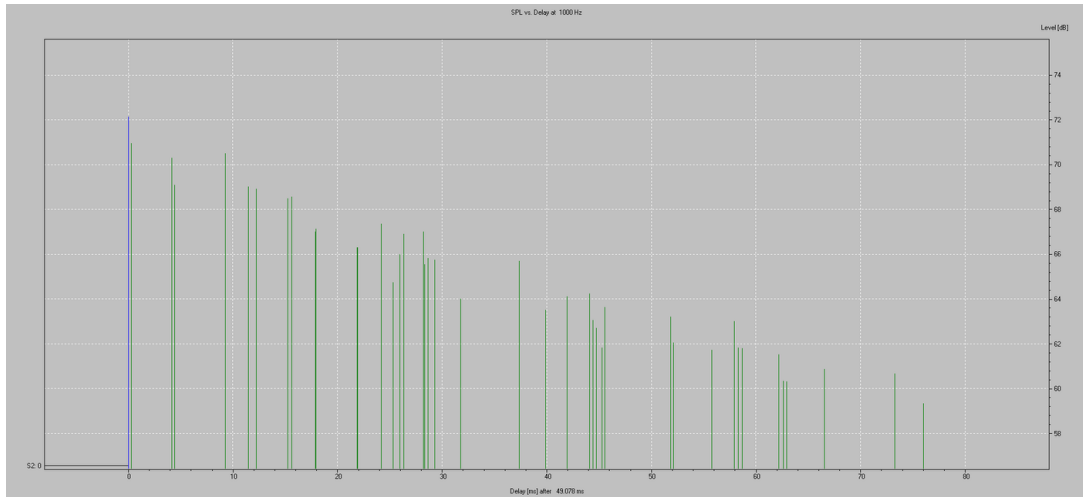


Figure 14: Echogram at seat 2, 1 kHz: SPL against delay after the 49.1 ms direct arrival, with no late reflection reaching Haas region D.

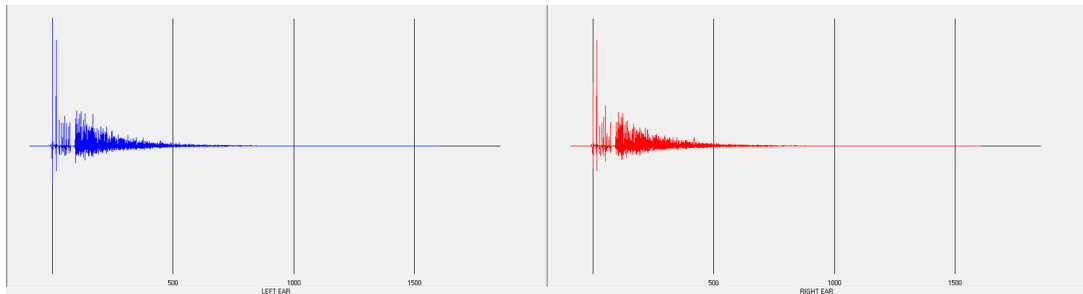


Figure 15: Binaural impulse response (left/right ear) generated for seat 2 with the EASE Ears module and used for auralisation.

6 Signal Processing in MATLAB

6.1 5-Band Parametric Equalizer

5-band parametric equalizer programmed entirely in MATLAB using object-oriented programming, focused on precise audio-signal manipulation and interactive visualisation of the frequency response.

- **Filters (Biquad):** second-order IIR filters (Direct Form II) for Peaking, High-Pass and Low-Pass stages; real-time computation of the complex transfer function $H(z)$ to derive the magnitude response.
- **Audio engine:** optimised processing loop guaranteeing smooth coefficient updates, preventing digital artefacts during dynamic manipulation of Gain, Freq and Q.
- **Analysis & UI:** interface (*uifigure*) built 100 % in code with logarithmic 20 Hz–20 kHz mapping on a decibel scale, plus preset system (Flat/Rock/Jazz/Vocal/Bass/Classical) and spectrogram and pole-zero analysis views.
- **Repository:** [GitHub – Parametric Equalizer](#)



Figure 16: Parametric EQ during playback: draggable band handles over the live input spectrum.

7 Machine Learning & Other Projects

7.1 QTGNN v2 — Quantum Temporal Graph Neural Network

Hybrid classical–quantum model for next-day stock price prediction, documented in a full technical report.

- **Architecture:** GRU for temporal encoding of price and volume, FinBERT transformer for real-time financial news sentiment, and a *Quantum Graph Attention Network* propagating information over a stock-correlation graph through a 4-qubit variational quantum circuit.
- **Data & validation:** 10 S&P 500 stocks across 5 sectors, ~500 days of history (OHLCV via yfinance) plus macroeconomic indicators, validated with walk-forward cross-validation.
- **Repository:** [GitHub – QTGNN-Finance](#)

7.2 AetherMap — Visual Blast-Radius Review (GitLab Transcend Hackathon, June 2026)

Agent for the GitLab Duo Agent Platform that turns the *Orbit Knowledge Graph* into a blast-radius review of every merge request.

- **Analysis engine:** maps changed hunks to the definitions that contain them and walks the call graph backwards (CALLS/EXTENDS edges) to find everything transitively impacted; scores risk LOW to CRITICAL, flags the most widely-called “fragile cores” and suggests reviewers.
- **Product:** Markdown report posted as an MR comment plus an interactive force-directed impact map; packaged as an Agent Skill with an MR-triggered CI flow. 100 % standard-library Python, MIT licence, unit-test suite.
- **Repository:** [GitHub – AetherMap](#)

7.3 Portfolio Website

Personal portfolio site with component-driven development in React (JSX) and Tailwind CSS, from UX wireframes to the optimised production export. **Repository:** [GitHub – Portfolio Website](#)

8 Technical Skills

■ Languages & paradigms:

- Modern C++ (C++17/20), Python, MATLAB and JavaScript (React, Electron).
- Object-oriented programming: modular architecture, structured class design and strict encapsulation.

■ Audio software development:

- **JUCE 8:** full plugin development in industry-standard formats (VST3, Standalone) and binary publishing.
- **Real-time DSP:** IIR/FIR filters, FFT, dynamics control, wavetable/additive synthesis, oversampling and algorithmic reverberation.
- Lock-free GUI/audio communication: atomics, FIFOs and ring buffers.

■ Room acoustics:

- EASE 4.4: 3D room modelling, material assignment, Eyring/Sabine reverberation-time prediction, ray tracing and mirror-image impacts.
- Objective evaluation to UNE-EN ISO 3382-1: echograms and Haas analysis, STI/RASTI mapping and binaural auralisation.

■ Computer vision & ML:

- OpenCV, MediaPipe (hand tracking), YOLOv8 and librosa (beat analysis).
- Neural networks (GRU, transformers, GNN) and quantum machine learning with variational circuits.

■ Robotics, embedded & hardware:

- ROS2 (Jazzy), MAVLink, LiDAR SLAM (Point-LIO) and rosbridge integration.
- ESP32 (I²S, I²C, SPI), circuit design in Fritzing and parametric modelling with the Autodesk Fusion 360 Python API.

■ Tools & environments:

- Microsoft Visual Studio 2022 (primary environment), MATLAB Editor.
- Git and GitHub (repositories and Releases), Vercel deployment, GitLab CI, L^AT_EX.

- Third-party API integration: LALAL.AI, Musixmatch, GitLab Orbit.